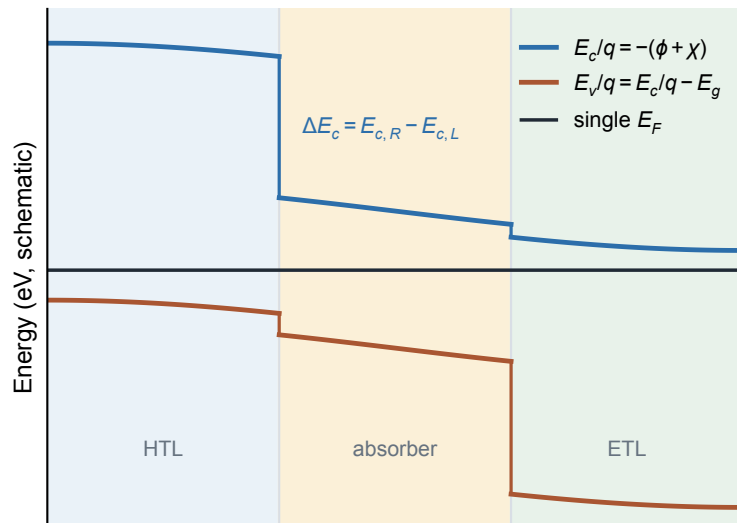
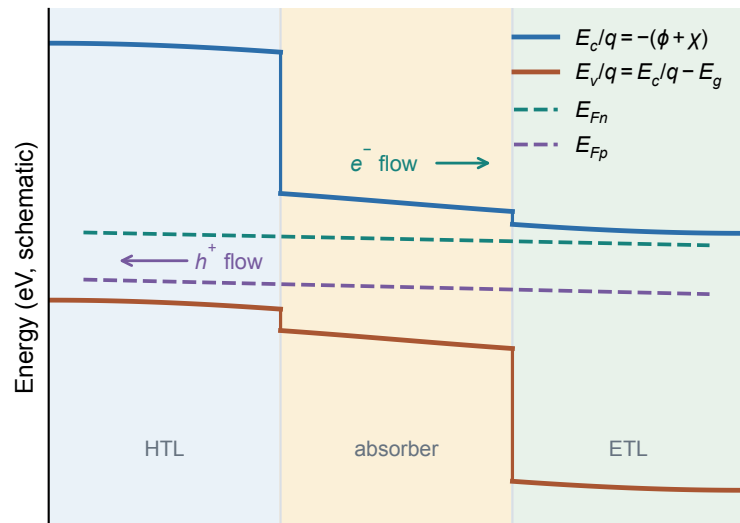


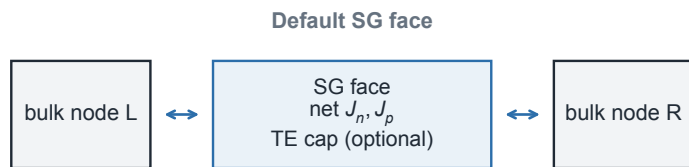
(a) Dark equilibrium with band bending



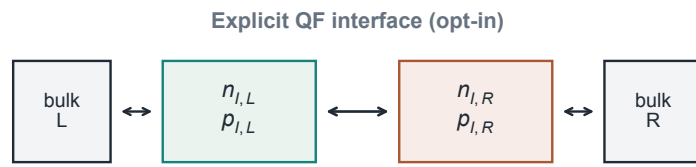
(b) Illuminated operation with band bending



(c) Abrupt-interface transport choices



bidirectional drift-diffusion face flux



$F_{\text{bulk}} + F_{\text{cross}} + F_{\text{SRH}} = 0$
zero thickness; shared trap occupancy